Aaditya Desai

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Education

San Jose State University — Expected May 2028
B.S. Computer Engineering, Minor in Artificial Intelligence | GPA: 3.8 | Dean's Scholar — *San Jose, CA*

- **Coursework:** Computer Architecture, x86 Assembly, Data Structures and Algorithms (in progress)

De Anza College — Mar 2026
Certificate of Achievement, Programming in C/C++, with Honors | GPA: 3.8 | Dean's List — *Cupertino, CA*

Experience

Infineon Technologies (in collaboration with De Anza College) — Apr 2026 – Jun 2026
Research Intern and Technical Lead, VisionAssist | github.com/aadityad12/Vision-Assist — *Cupertino, CA*

- Led software on a 5-person team building an offline navigation aid for blind users on a Raspberry Pi 5: YOLOv8n detection at about 200 ms per frame on CPU, distance estimation, and spoken alerts via a headless systemd service.
- Evaluated the sponsor's 60 GHz radar, traced its phantom close-range readings to antenna leakage and floor reflections, showed camera-radar fusion needed 3D localization beyond the project's scope, and delivered the findings to Infineon.

San Jose State University — Aug 2026 – Present
Teaching Assistant, ENGR 10: Introduction to Engineering — *San Jose, CA*

- Run two weekly labs of 30 students who solder IR receiver boards and program VEX robots in C++ or RobotC.

De Anza College — Jan 2025 – Dec 2025
Teaching Assistant and Computer Science Tutor — *Cupertino, CA*

- Taught C++, x86 assembly, and data structures as TA for three courses and CS lab tutor; traced bugs to root cause.

Projects

Interpreter and ARM64 JIT Compiler (Rung) — Sep 2026 – Oct 2026
C++20, ARM64, CMake, ASan/UBSan/TSan, libFuzzer, GitHub Actions — github.com/aadityad12/rung

- Built one small language five ways (tree-walker, stack VM, register VM, baseline ARM64 JIT, background-thread JIT) on a shared front end and runtime to measure what each speed-up is worth; all engines must print byte-identical output.
- Wrote an ARM64 encoder checked byte-for-byte against LLVM; the JIT ran a hot loop 22.6x faster than the tree-walker.
- Ran every engine through a conformance suite, sanitizer builds, and a fuzzer in CI on three platforms; root-caused a JIT crash in about 5% of Linux ARM64 runs to UBSan reading 8 bytes before the call target.

Reversible Context Manager for AI Coding Agents (Accordion) — Jun 2026
TypeScript, SvelteKit, Tauri, Hugging Face Transformers — github.com/a-Fig/accordion

- Folds stale context out of a coding agent's view instead of deleting it; won the Sponsor Track at the UC Berkeley AI Hackathon 2026 with a team of four, and the open-source repo has 240+ GitHub stars.
- Built the folding policy's health and evaluation layer, including a harness that compares folding against naive compaction, and ported the policy onto Accordion's fold, replace, and group contract.
- Added a cross-encoder rerank stage (MiniLM) to the unfold shortlist and a dashboard showing every fold decision.

Least-Privilege Agent Pipeline for Open-Source Work (OSS Scout) — Sep 2026 – Oct 2026
Python, GitHub Actions, Claude Code, Cloudflare Workers, pytest — github.com/aadityad12/oss-scout

- Built a nightly scan in plain Python that finds open bug and beginner issues and scores each project by how it treats outside contributors: merge rate, time to merge, and time to first maintainer reply.
- Built the review path: a Claude agent prepares at most one tested draft fix per night for a private dashboard behind Cloudflare Access, and nothing posts without a one-tap approval.
- Kept the unattended agent unable to write to GitHub: a tested hook blocks its writes and the only write token lives in a separate private repo; about 280 pytest tests cover the pipeline.

Technical Skills

Languages: C++, C, Python, Kotlin, TypeScript, JavaScript, SQL, ARM64 and x86 assembly
Systems and Testing: Linux, CMake, Docker, GitHub Actions, Git, ASan/UBSan/TSan, libFuzzer, pytest
Frameworks: FastAPI, Flask, React, SvelteKit, Jetpack Compose, Firebase, Cloudflare Workers
AI and Hardware: PyTorch, Hugging Face Transformers, YOLOv8, OpenCV, Claude Code, Raspberry Pi, soldering

Awards and Leadership

Awards: ICPC 2025 Regionals, top 10 at site (Division 2); DA Hacks 3.5, 2nd place
Leadership: Co-Founder and Co-President, Applied Data Analytics Club, De Anza College (2025); grew it to 54 members
Certifications: AWS Certified Cloud Practitioner; Claude Code in Action and Model Context Protocol (Anthropic)